

Where Prices Come From: The Interaction of Supply and Demand

ECON 101 – Macroeconomics

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Chapter Outline

- **3.1 The Demand Side of the Market**
- 3.2 The Supply Side of the Market
- 3.3 Market Equilibrium: Putting Buyers and Sellers Together
- 3.4 The Effect of Demand and Supply Shifts on Equilibrium

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3.1 Learning Objective

Goal

Discuss the variables that influence demand.

- We begin by investigating how **buyers** behave.
- **Market demand:** the demand by all consumers of a given good or service.
- Key tools: demand schedule and demand curve.
- Running example: the market for **Tim Hortons medium coffee** in Canada.

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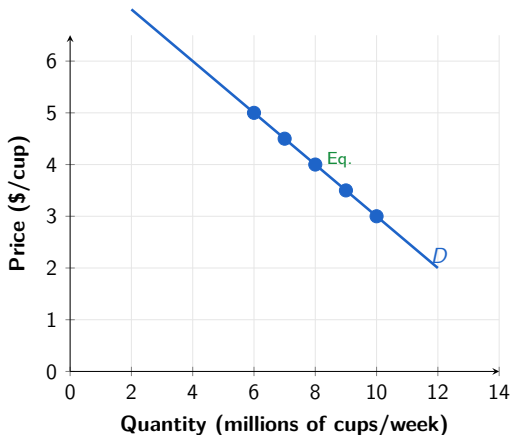
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Figure 3.1 – Demand Schedule and Demand Curve

Tim Hortons Medium Coffee Demand Schedule

Price (\$/cup)	Qty Demanded (M cups/week)
\$3.00	10
\$3.50	9
\$4.00	8
\$4.50	7
\$5.00	6

Source: Illustrative; based on
Statistics Canada CPI data



Demand Schedule and Demand Curve: Key Concepts

- **Demand schedule:** a table showing the relationship between the **price** and the **quantity demanded**.
- **Demand curve:** a curve showing the same relationship graphically.
- When drawing the demand curve, we hold all other factors constant — **ceteris paribus**.
- **Ceteris paribus:** “all else equal” — when analyzing price and quantity demanded, other variables are held constant.

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The Law of Demand

- **Quantity demanded:** the amount of a good a consumer is willing and able to purchase at a given price.
- **Law of demand:**
 - Holding everything else constant, when the price of a product falls, the quantity demanded increases.
 - When the price rises, the quantity demanded decreases.
- **Canadian data:** When Tim Hortons raised medium coffee prices by $\approx 10\%$ in 2022–2023, same-store transaction counts fell in the short run (Restaurant Brands International earnings, 2023).

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What Explains the Law of Demand?

- When the price of a good falls, two key effects occur:
 - **Substitution effect:** consumers substitute toward the good whose price has fallen.
 - **Income effect:** consumers have more real purchasing power, which raises quantity demanded.
- **Substitution effect:** the change in quantity demanded resulting from a change in price relative to other goods, *holding purchasing power constant*.
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Canadian Example: Substitution and Income Effects

- Suppose Tim Hortons medium coffee falls from \$4.00 to \$3.00.
- **Substitution effect:**
 - Coffee is now cheaper relative to Starbucks lattes, Red Bull, or tea — consumers substitute toward Tim Hortons coffee.
- **Income effect:**
 - A daily coffee drinker saves $\approx \$365/\text{year}$ — that extra real income allows more cups per week.

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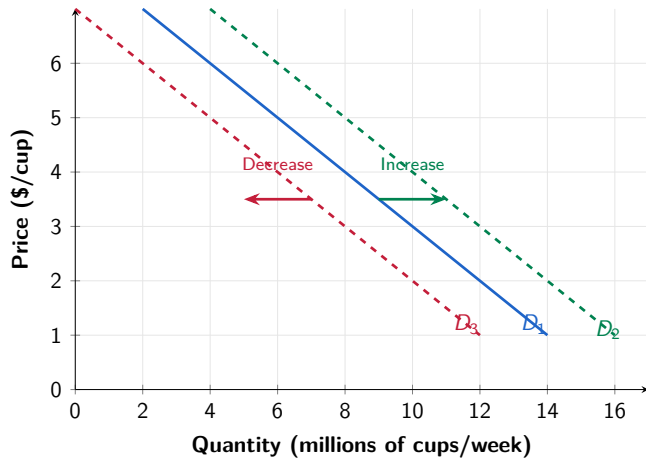
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Figure 3.2 – Shifting the Demand Curve



Shifts in Demand: Key Distinction

- A change in something **other than price** causes the **entire demand curve to shift**.
- $D_1 \rightarrow D_2$ (right): an **increase in demand** — more coffee demanded at every price.
- $D_1 \rightarrow D_3$ (left): a **decrease in demand** — less coffee demanded at every price.
- The quantity demanded changes at **every possible price** when the curve shifts.

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- Prices of related goods (substitutes and complements)
- Tastes
- Population and demographics
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Changes in Income: Normal vs. Inferior Goods

- **Normal good:** demand increases as income rises (e.g., restaurant meals, new cars, Starbucks lattes).
- **Inferior good:** demand decreases as income rises (e.g., instant noodles, transit bus passes).
- **Canadian context:**
 - Tim Hortons brewed coffee may behave as a **normal good** overall, but an **inferior good** relative to specialty cafes — as incomes rose in 2010–2019, Starbucks expanded in Canada while Tims' transaction growth slowed (Restaurants Canada, 2019).

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Changes in the Price of Related Goods

- **Substitutes:** goods that can be used for the same purpose.
 - Examples: Tim Hortons coffee and Starbucks latte; Ford F-150 and RAM 1500; WestJet and Air Canada.
- **Complements:** goods used together.
 - Examples: coffee and cream; hot dogs and buns; smartphones and data plans.
- If the price of a Starbucks latte rises, demand for Tim Hortons coffee **increases**.
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Changes in Tastes, Population, and Expectations

- **Tastes:** growing health concerns reduce demand for sugary drinks; specialty coffee trends shift demand toward dark roast and espresso.
- **Population and demographics:** Canada's population grew from 38M to 41M between 2021 and 2024 (Statistics Canada, 2024) — more Canadians means more coffee drinkers.
- **Expectations:** if consumers expect coffee prices to jump next month (e.g., due to a drought in Brazil), they increase demand **today**.

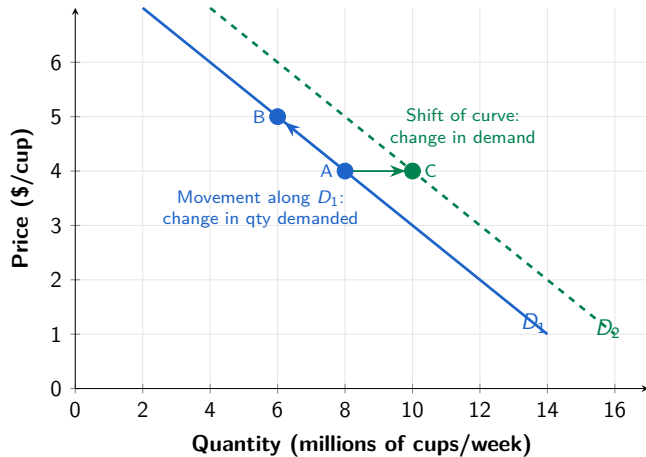
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Figure 3.3 – Change in Demand vs. Change in Quantity Demanded



Change in Demand vs. Change in Quantity Demanded

- A change in the **price of the product** causes a **movement along** the demand curve — this is a **change in quantity demanded**.
- Any other change affecting demand causes the **entire demand curve to shift** — this is a **change in demand**.
- **Common mistake:** saying “the decrease in price causes demand to increase.” The decrease in price causes a movement along the curve, not a shift.

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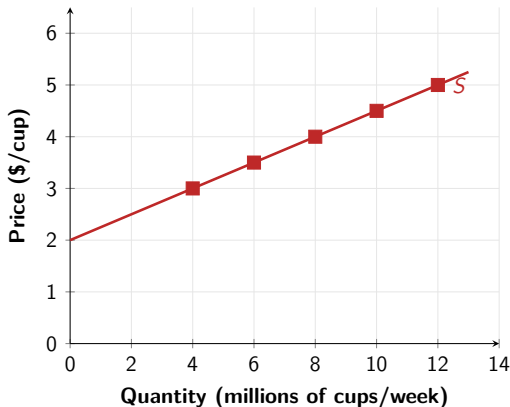
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Figure 3.4 – Supply Schedule and Supply Curve

Tim Hortons Medium Coffee Supply Schedule

Price (\$/cup)	Qty Supplied (M cups/week)
\$3.00	4
\$3.50	6
\$4.00	8
\$4.50	10
\$5.00	12

Source: Illustrative; based on
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The Law of Supply

- **Quantity supplied:** the amount of a good a firm is willing and able to supply at a given price.
- **Law of supply:**
 - Holding everything else constant, increases in price cause **increases** in quantity supplied.
 - Decreases in price cause **decreases** in quantity supplied.
- **Why?** Higher prices make production more profitable, attracting more supply.
- **Canadian context:** When canola prices surged in 2021–2022, Prairie farmers allocated more acres to canola production (Statistics Canada, Field Crop Reporting Series).

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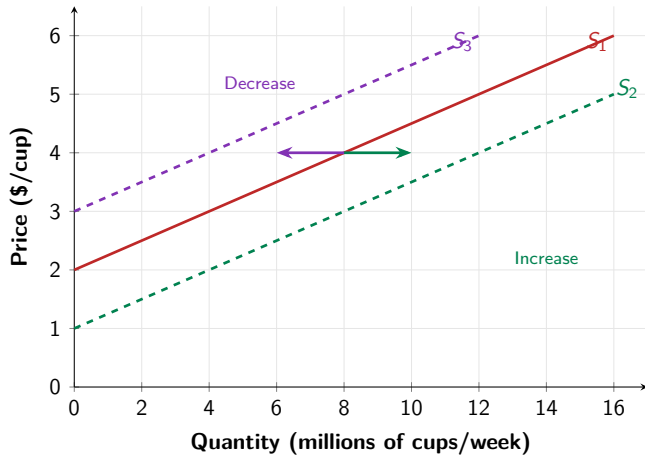
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Figure 3.5 – Shifting the Supply Curve



Variables That Shift Market Supply

- **Prices of inputs** (labour, coffee beans, milk, paper cups)
- Technological change (brewing equipment efficiency)
- Prices of substitutes in production
- Number of firms in the market
- Expected future prices

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Changes in Prices of Inputs

- **Inputs:** anything used in production — for coffee: arabica beans, milk, labour, paper cups, electricity.
- An **increase** in input prices decreases profitability $\Rightarrow$ **decrease in supply** (curve shifts left).
- **Canadian example:** Global arabica coffee bean prices rose $\approx 70\%$ in 2024–2025 due to droughts in Brazil, raising costs for all Canadian coffee chains and reducing supply (ICO Coffee Report, 2025).

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Technological Change and Other Supply Shifters

- **Technological change:** new, more efficient brewing systems $\Rightarrow$ **increase in supply.**
- **Substitutes in production:** an Ontario farmer can plant corn or soybeans.
 - If soybean prices rise, farmers shift to soybeans $\Rightarrow$ **decrease in corn supply.**
- **Complements in production:** beef and leather (more cattle farming raises supply of both).
- **Number of firms:** more coffee shops entering the market $\Rightarrow$ **increase in supply.**
- **Expected future prices:** if coffee prices expected to rise, firms may withhold supply now.

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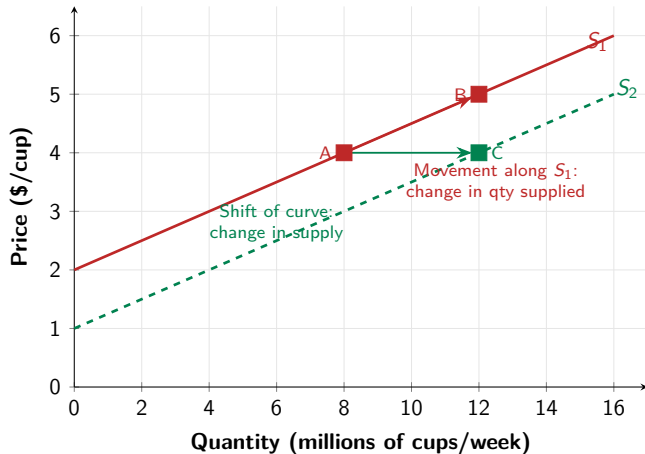
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Figure 3.6 – Change in Supply vs. Change in Quantity Supplied



3.3 Learning Objective

Goal

Use a graph to illustrate market equilibrium.

- **Market equilibrium:** quantity demanded equals quantity supplied.
- In perfectly competitive markets, this is called a **competitive market equilibrium**.
- The equilibrium price is the price that *clears* the market.

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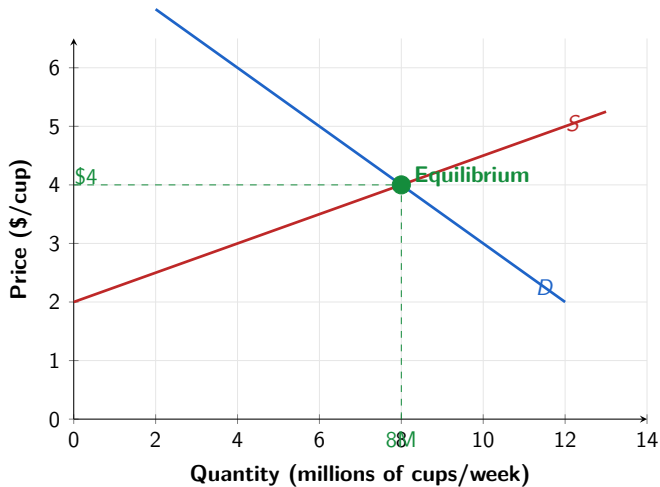
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Figure 3.7 – Market Equilibrium: Tim Hortons Coffee



Reading the Equilibrium

- At a price of **\$4.00/cup**:
 - Consumers want to buy **8 million** cups/week.
 - Producers want to sell **8 million** cups/week.
- The **equilibrium price** is \$4.00/cup; the **equilibrium quantity** is 8 million cups/week.
- At any other price, the market is out of balance — either a surplus or a shortage exists.

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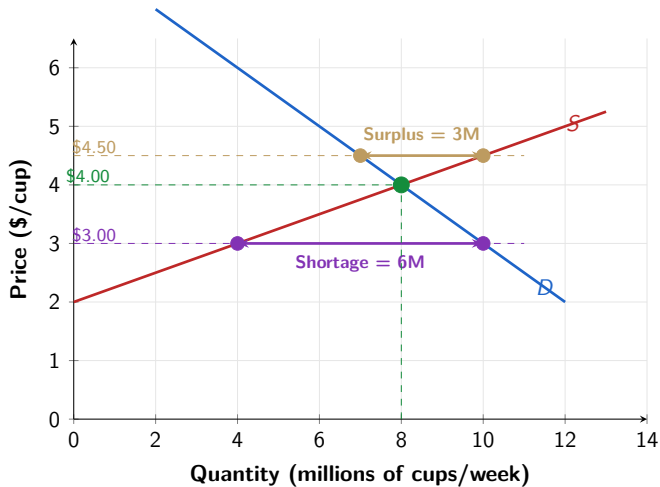
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Figure 3.8 – Surplus and Shortage



Surplus and Shortage

- **Surplus** ($P = \$4.50$): $Q_s = 10M > Q_d = 7M$ — surplus of 3M cups/week.
 - Sellers compete among themselves $\Rightarrow$ price is bid **down** toward \$4.00.
- **Shortage** ($P = \$3.00$): $Q_d = 10M > Q_s = 4M$ — shortage of 6M cups/week.
 - Buyers compete among themselves $\Rightarrow$ price is bid **up** toward \$4.00.
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Use demand and supply graphs to predict changes in prices and quantities.

- We typically observe the current price and quantity, but do *not* observe the whole curves.
- The model predicts **directional** changes in price and quantity when curves shift.

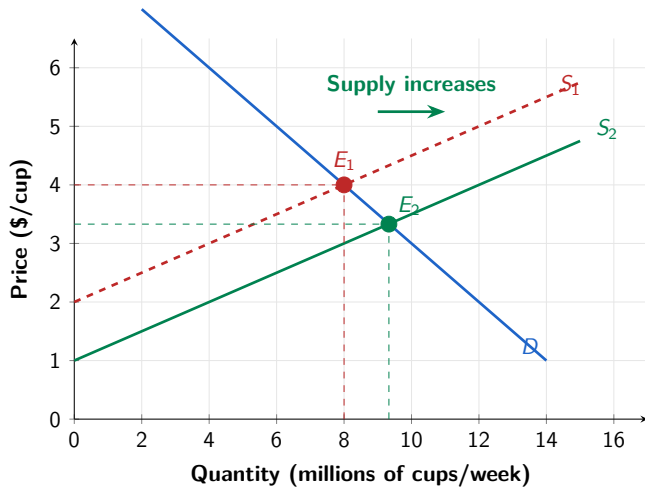
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Figure 3.9 – Effect of an Increase in Supply



Effect of an Increase in Supply

- New coffee chains enter the Canadian market (e.g., Dutch Bros. expanding into Canada).
- Supply shifts **right** from S_1 to S_2 .
- **Result:**
 - Equilibrium price falls (\$4.00 $\rightarrow$ \$3.33).
 - Equilibrium quantity rises (8M $\rightarrow$ 9.3M cups/week).
- We only predict *directions* — the exact magnitude requires knowing the full curves.

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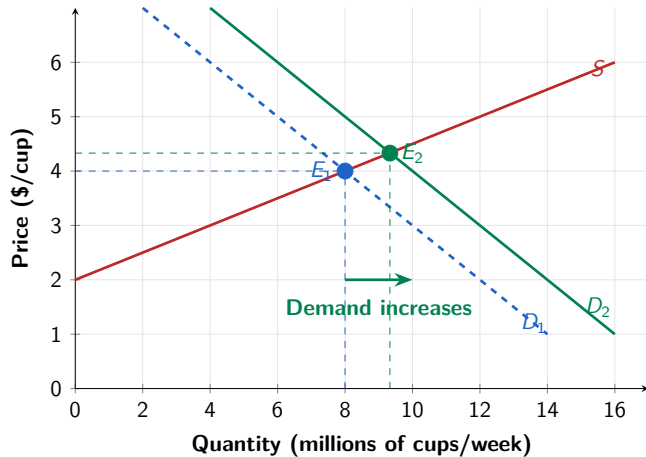
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Figure 3.10 – Effect of an Increase in Demand



Effect of an Increase in Demand

- Canada's population grew by ≈ 3 million in 2022–2024 (Statistics Canada) — more coffee drinkers.
- Demand shifts **right** from D_1 to D_2 .
- **Result:**
 - Equilibrium price rises (\$4.00 $\rightarrow$ \$4.33).
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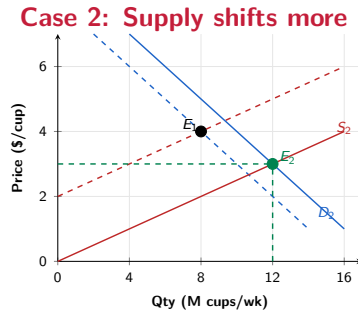
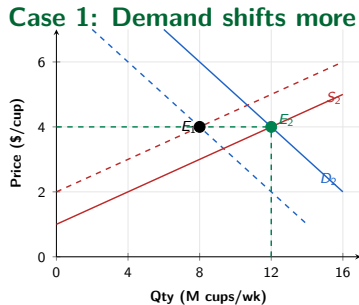
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Table 3.3 – How Shifts Affect Equilibrium P and Q

Change	Eq. Price P	Eq. Quantity Q
Demand $\uparrow$, supply unchanged	Rises	Rises
Demand $\downarrow$, supply unchanged	Falls	Falls
Supply $\uparrow$, demand unchanged	Falls	Rises
Supply $\downarrow$, demand unchanged	Rises	Falls
Demand $\uparrow$ and supply $\downarrow$	Rises	?
Demand $\downarrow$ and supply $\uparrow$	Falls	?
Demand $\uparrow$ and supply $\uparrow$	?	Rises
Demand $\downarrow$ and supply $\downarrow$	?	Falls

Figure 3.11 – Both Demand and Supply Shift



Both cases: quantity **rises**. Price: **ambiguous** — depends on which shift is larger.

Shifts of Curves vs. Movements Along Curves

- Suppose an **increase in supply** occurs.
 - Equilibrium quantity increases; equilibrium price decreases.
- It is tempting to say “the decrease in price causes an increase in demand” — **incorrect**.
- The decrease in price causes a **movement along the demand curve**, not a shift.
- The demand curve already describes how much consumers want to buy at **any given price**.
- **Canadian example:** When new discount airlines entered the YVR–YYZ route, fares fell and more passengers flew — that is a movement along demand, not a demand shift.

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Things to Remember

- A **change in price** causes a movement **along** the demand or supply curve.
- Any **other change** shifts the entire curve.
- In equilibrium, $Q_d = Q_s$; surpluses push price **down**; shortages push price **up**.
- When both curves shift, one effect is **ambiguous** — need to know relative magnitudes.
- Markets coordinate millions of buyers and sellers through prices — no central planner required.

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Chapter Summary

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- **Supply** follows the law of supply. Shifts when input prices, technology, production substitutes/complements, number of firms, or expected future prices change.
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Questions?

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